



LAB REPORT 02

JAVA

Submitted by,

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Language II (Java) Lab
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Date of submission : 09.09.2026

A line drawing of a pen writing the name 'Bayazid' in a cursive script. The pen is positioned diagonally, and the signature is written on a horizontal line.

Part 01

- **Encapsulation and Data Hiding**
- **Private data members in a Student class**

How to do it -

1. Create a Product class for storing product information.
2. Declare product name, ID, and price as private.
3. Private variables cannot be directly accessed from outside the class.
4. Create a public method to set the product information.

This method takes the values and stores them inside the private variables.

So instead of directly changing the data, we use a controlled method to access it.

That is the basic idea of encapsulation.

5. Create another public method to display the information.

```
// Topic 1: Encapsulation and Data Hiding
// Topic 2: Private Data Members

class Product {

    // Private data members
    private String productName;
    private int productId;
    private double price;

    // Method to set product information
    public void setProduct(String name, int id, double price) {
        productName = name;
        productId = id;
        this.price = price;
    }

    // Method to display product information
    public void displayProduct() {
        System.out.println("Product ID : " + productId);
        System.out.println("Product Name : " + productName);
        System.out.println("Price : " + price);
    }
}
```

Part 02

- **Getter and Setter methods**
- **Use of this keyword**

Concept of Get and Set

We learned from the previous chapter that private variables can only be accessed within the same class (an outside class has no access to it). However, it is possible to access them if we provide **public get** and **set** methods.

The **get** method returns the variable value, and the **set** method sets the value.

Examples

```
public class Person {  
    private String name; // private = restricted access  
  
    // Getter  
    public String getName() {  
        return name;  
    }  
  
    // Setter  
    public void setName(String newName) {  
        this.name = newName;  
    }  
}
```

How to do it -

1. Create a **Customer** class for the SuperShop.
2. Keep customer information private.
3. Create setter methods to store or change information.
4. Create getter methods to read the private information.
5. Use the **this** keyword to refer to the object's own variables.

```

// Topic 3: Getter and Setter Methods
// Topic 4: Use of this Keyword

class Customer {

    // Private data members
    private String name;
    private String phone;
    private double purchase;

    // Setter methods
    public void setName(String name) {
        this.name = name;
    }

    public void setPhone(String phone) {
        this.phone = phone;
    }

    public void setPurchase(double purchase) {
        this.purchase = purchase;
    }

    // Getter methods
    public String getName() {
        return name;
    }

    public String getPhone() {
        return phone;
    }

    public double getPurchase() {
        return purchase;
    }

    // Display customer information
    public void displayCustomer() {
        System.out.println("Customer Name : " + getName());
        System.out.println("Phone Number : " + getPhone());
        System.out.println("Purchase : " + getPurchase());
    }
}

```

why this Keyword ?

`this.name = name;`

`this.name` → object's instance variable

`name` → method parameter

So we use `this` to put the value received by the method into the object's `name` variable.

Part 03

- **Constructors with encapsulation**
- **Passing objects as an array and performing operations using getter/setter methods**

How to do it -

1. Create an Item class for products available in the SuperShop.
2. Keep all product information private.
3. Create a constructor to initialize the product when an object is created.
4. Create getter and setter methods for controlled access.
5. Create an array containing multiple Item objects.
6. Use a loop and getter methods to calculate the total value of all products.

```
// Topic 5: Constructors with Encapsulation
// Topic 6: Passing Objects as an Array and Operations using Getter/Setter

class Item {

    // Private data members
    private int id;
    private String name;
    private double price;
    private int quantity;

    // Constructor
    public Item(int id, String name, double price, int quantity) {
        this.id = id;
        this.name = name;
        this.price = price;
        this.quantity = quantity;
    }
}
```

```
// Getter methods
public int getId() {
    return id;
}

public String getName() {
    return name;
}

public double getPrice() {
    return price;
}

public int getQuantity() {
    return quantity;
}

// Setter method
public void setQuantity(int quantity) {
    this.quantity = quantity;
}

// Calculate total price
public double getTotalPrice() {
    return price * quantity;
}

// Display item information
public void displayItem() {
    System.out.println(
        id + " | " +
        name + " | " +
        price + " | " +
        quantity + " | " +
        getTotalPrice()
    );
}
}
```

Part 04

- **The Main Body**
- **Creating objects and calling methods from all three programs**

```
// Main Program
// Creating objects and calling methods from all three programs

public class Main {

    public static void main(String[] args) {

        //=>Calling Product

        System.out.println("\n=>Product Detail");

        Product p1 = new Product();

        p1.setProduct("Rice", 101, 750.00);

        p1.displayProduct();

        //=>Calling Customer

        System.out.println("\n=>Customer Detail");

        Customer c1 = new Customer();

        c1.setName("Ayon Mia");
        c1.setPhone("01700000000");
        c1.setPurchase(2500.00);

        c1.displayCustomer();
    }
}
```

//=>Calling Item

```
System.out.println("\n=>Item Detail");

// Creating objects using constructor
Item item1 = new Item(101, "Rice", 750.00, 5);
Item item2 = new Item(102, "Oil", 180.00, 10);
Item item3 = new Item(103, "Sugar", 120.00, 8);

// Passing objects as an array
Item[] items = {item1, item2, item3};

System.out.println("ID | Name | Price | Quantity | Total");
System.out.println("-----");

// Display all items
for (Item item : items) {
    item.displayItem();
}

// Calculate total inventory value
double total = 0;

for (Item item : items) {
    total = total + item.getTotalPrice();
}

System.out.println("\nTotal Inventory Value: " + total + "\n");

// Updating quantity using setter
item2.setQuantity(15);

System.out.println(
    "\nUpdated Oil Quantity: "
    + item2.getQuantity()
);

System.out.println(
    "Updated Oil Total: "
    + item2.getTotalPrice()
);
}
}
```

Output-

=>Product Detail

Product ID : 101

Product Name : Rice

Price : 750.0

=>Customer Detail

Customer Name : Ayon Mia

Phone Number : 01700000000

Purchase : 2500.0

=>Item Detail

ID	Name	Price	Quantity	Total
----	------	-------	----------	-------

101	Rice	750.0	5	3750.0
-----	------	-------	---	--------

102	Oil	180.0	10	1800.0
-----	-----	-------	----	--------

103	Sugar	120.0	8	960.0
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Total Inventory Value: 6510.0

Updated Oil Quantity: 15

Updated Oil Total: 2700.0

Resources



Prepared by
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